

# JAMES JAW

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## EDUCATION

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| <b>University of Wisconsin-Madison</b><br><i>M.S. in Computer Science</i>             | Sep 2023 – Dec 2025<br><i>GPA: 3.9</i> |
| <b>National Tsing Hua University</b><br><i>Post-Baccalaureate in Computer Science</i> | Sep 2020 – Jan 2022<br><i>GPA: 3.9</i> |
| <b>Tunghai University</b><br><i>B.B.A. in Business Administration</i>                 | Sep 2014 – Jul 2018                    |

## SKILLS

**Languages:** Python, Golang, Java, C++, JavaScript, TypeScript, HTML/CSS  
**Frameworks:** Django, Spring, React, Node.js, gRPC, REST APIs, PostgreSQL, MySQL  
**Infrastructure:** Docker, Kubernetes, GitHub Actions, AWS, Linux

## EXPERIENCE

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| <b>Open Source – Contributor</b><br>KubeRay, Apache Airflow – <i>Golang, Python, TypeScript, Docker, Kubernetes</i> <ul style="list-style-type: none"><li>– Diagnosed a KubeRay worker-startup bottleneck: right-sized the wait-gcs-ready init container’s CPU/memory to match the Ray container, cutting up to <b>55s</b> of autoscaling latency per new worker</li><li>– Resolved a flaky e2e test caused by a concurrent-update race condition and modernized lint rules across the operator codebase; both merged upstream</li><li>– Contributed 5 merged PRs to Apache Airflow, including a React rendering fix for duplicate-key list reconciliation in the web UI and a declarative YAML migration of provider connection metadata across 3 provider packages</li></ul> | Feb 2026 – Present  |
| <b>Digital Aid Seattle – Software Engineer</b><br>Scheduling Platform – <i>TypeScript, Next.js, Supabase (Postgres/Auth/Edge Functions)</i> <ul style="list-style-type: none"><li>– Built a full-stack scheduling tool that automates student grouping via timezone-aware availability matching, replacing manual spreadsheet workflows for multiple instructors</li><li>– Designed multi-tenant data isolation with Supabase Row Level Security, enabling instructors to independently manage class data on a shared platform</li></ul>                                                                                                                                                                                                                                       | May 2026 – Present  |
| <b>IBM – Software Engineer Intern</b><br>Middleware Modernization – <i>C++, Shell, Docker, Kubernetes, Argo CD, Grafana</i> <ul style="list-style-type: none"><li>– Built a C++ middleware integrating legacy CORBA APIs with modern deployment infrastructure, automating a previously manual validation workflow (<b>~60% faster</b>)</li><li>– Migrated legacy C++ services to Kubernetes, replacing manual deployments with Blue-Green GitOps releases</li><li>– Built team-friendly k6 gRPC load-testing tooling to simulate <b>1,000+</b> concurrent users and expose bottlenecks</li></ul>                                                                                                                                                                              | Jul 2025 – Aug 2025 |

## PROJECTS

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| <b>Video Chat Platform</b> – <i>Golang, Fiber, TypeScript, React, WebRTC, Postgres</i> <ul style="list-style-type: none"><li>– Shipped BuckyChat, a campus-only video chat platform for verified UW–Madison students, as the sole engineer from zero to production</li><li>– Architected peer-to-peer WebRTC media over a lightweight Go signaling backend with Cloudflare TURN fallback; k6 load tests sustained <b>200</b> concurrent WebSocket sessions and <b>27,000+</b> connections at <b>100%</b> handshake success</li><li>– Enforced campus-only access via @wisc.edu domain allowlisting at signup and JWT-validated WebSocket sessions, with per-user rate limiting to curb abuse</li></ul> | 2026 |
| <b>LLM-Based Database Schema Name Expansion</b> – <i>Python, OpenAI API, spaCy</i> <ul style="list-style-type: none"><li>– Co-built an LLM pipeline that expands cryptic database column names (e.g., ModDate → Modified Date), achieving <b>76–88% accuracy</b> across 4 relational datasets</li><li>– Designed the evaluation harness that benchmarked all prompt variants: scoring LLM predictions against curated gold labels with precision/recall, synonym, and fuzzy matching</li></ul>                                                                                                                                                                                                         | 2025 |